B. TECH-ME-III-II Sem

23A03605a	APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS (Professional Elective-III)	L	T	P	C
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Course objectives: The objectives of the course are to	
1	Foundation in numerical techniques, and finite element methods for solving partial differential equations under various boundary conditions.
2	Solid understanding of numerical methods for solving time-dependent partial differential equations, with emphasis on stability and accuracy analysis.
3	To introduce students to numerical formulations for incompressible and compressible viscous flows using finite difference and advanced computational techniques, enabling them to model and analyze fluid flow problems governed by the Euler and Navier-Stokes equations.
4	Knowledge and skills to apply the finite volume method using finite difference formulations for solving two- and three-dimensional fluid flow and heat transfer problems.
5	Understand the concepts of linear fluid flow problems, steady state problems and transient problems.

COURSE OUTCOMES On successful completion of this course the student will be able to		
1	Formulate and solve partial differential equations using finite element methods, apply and analyze the stability, accuracy of explicit and implicit methods.	*L2,L3,L4*
2	Apply and analyze explicit and implicit numerical schemes, solve nonlinear and second-order PDEs, using appropriate numerical techniques.	L3,L4,L5
3	Create and solve incompressible and compressible viscous flow problems, apply appropriate boundary conditions.	L2,L5,L6
4	Develop and implement finite volume formulations based on finite difference methods, ensuring accurate and conservative solutions in CFD applications.	L1,L3,L6
5	Analyze and solve linear fluid flow problems, using appropriate numerical methods and interpret the physical significance of the computed results.	L1,L4,L5

UNIT I

Introduction and Solution methods

Introduction: Finite difference method, finite volume method, finite element method, governing equations and boundary conditions, Derivation of finite difference equations.

Solution methods: Solution methods of elliptical equations — finite difference formulations, interactive solution methods, direct method with Gaussian elimination. Parabolic equations-explicit schemes and Von Neumann stability analysis, implicit schemes, alternating direction implicit schemes, approximate factorization, fractional step methods, direct method with tridiagonal matrix algorithm.

UNIT II

Hyperbolic equations:

Hyperbolic equations:explicit schemes and Von Neumann stability analysis, implicit schemes, multi step methods, nonlinear problems, second order one-dimensional wave equations. Burgers equations: Explicit and implicit schemes, Runge-Kutta method.

UNIT III

Formulations Of Incompressible Viscous Flows

Formulations Of Incompressible Viscous Flows: Formulations of incompressible viscous flows by finite difference methods, pressure correction methods, vortex methods.

Treatment of compressible flows: potential equation, Euler equations, Navier-stokes system of equations, flow field-dependent variation methods, boundary conditions, example problems.

UNIT IV

Finite Volume Method:

Finite Volume Method Finite volume method via finite difference method, formulations for two and three-dimensional problems.

UNIT V

Standard Variational Methods:

Standard Variational Methods Linear fluid flow problems, steady state problems, Transient problems.

Textbooks:

1. T. J. C'hung, Computational fluid dynamics, Cambridge University press,2002.
2. John D. Anderson, Computational Fluid Dynamics: Basics with applications, Mc Graw Hill. 2017

Reference Books:

1. Frank Choriton, Text book of fluid dynamics, CBS Publishers & distributors, 1985.
2. Suhas V. Patankar, Numerical heat transfer and fluid flow, Hema shava Publishers corporation & Mc Graw Hill, 1990.
3. Muralidaran, Computational Fluid Flow and Heat Transfer, Narosa Publications, 2003.
4. Tapan K. Sengupta, Fundamentals of Computational Fluid Dynamics, Universities Press, 2004.
5. C. Pozrikidis, Introduction to Theoretical and Computational Fluid Dynamics, Oxford University press, 2/e, 2012.

Online Learning Resources:

- https://nptel.ac.in/courses/112107079
- https://www.youtube.com/watch?v=3QFT7pGx03I
- https://www.youtube.com/watch?v=t7jS7V_6TGQ
- https://nptel.ac.in/courses/112107080